

From Experimental Histories to Causal Surrogates: Data Reuse, Optimization, and Error Propagation

Research draft

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Abstract

A sequential experimental learner need not be the best model delivered after collection ends. We investigate whether retaining and refitting a completed experimental history improves the resulting causal surrogate, and which parts of the delivery procedure account for that change. A completed confirmation on one fixed deterministic emulator covers 12 acquisition histories: refitting all eligible paid observations reduces geometric mean snapped exact-level error to 0.183 times the online learner’s error (95% interval 0.102–0.327), with 11 histories improving. This comparison combines data reuse and additional optimization; it does not identify an architecture effect or an advantage at equal fitting compute. An exploratory attribution study on the same exposed grid gives continuous-error ratios of 0.551 versus a development-selected flat refit and 0.423 versus a matched-CPU flat fit, using fixed initialization zero. We give conditional statements for mechanism eligibility, deterministic error propagation and quantization sensitivity, with explicit counterexamples to monotonic improvement. An archived physical case study retains all eleven conditions: delivery beats rolling learning in 7 conditions but Fourier regression in 0. Independent-system comparisons remain necessary before submission. The present evidence supports a scoped post-collection delivery finding, without an acquisition or foundation-model claim.

1 Introduction

Experimental collection and final model delivery have different constraints. An online learner must update quickly and may discard observations to bound memory or computation. After collection, a separate fit can use retained observations, including paid probes and former validation rows. Any apparent benefit then has several possible explanations: a larger retained data set, more fitting compute, different optimization history, or the use of a known mechanism graph. Our aim is to separate these explanations.

The completed comparison supplies evidence for a delivery recipe on one emulator. The experimental program below is designed to test whether that recipe has a distinct causal-surrogate contribution. An ordinary flat refit explaining the result would narrow the contribution to delivery and experimental accounting. We will retain that conclusion rather than infer a more general architectural advantage from a favorable aggregate. Completed exploratory attribution supports carrying the structured recipe to a prospective test, while retaining its worsening history and limitations.

Current contributions and scope. We report a completed, receipt-backed comparison across repeated acquisition histories of one fixed system; derive conditions and limitations relevant to reuse and chain prediction; and report exploratory attribution that separates retained data from fitting

budget. An archived physical case study establishes a boundary against strong domain controls; independent-system results are pending. Neither causal structure discovery nor learned experiment selection is a current contribution. No foundation model is used in the delivery experiment.

2 Problem and related work

We assume a known directed acyclic graph. A surrogate predicts each nonroot node from its measured parents during fitting and its predicted parents during free-running deployment. Correct interventions replace the relevant prediction with its clamped value. A roots-to-target flat predictor supplies a simpler prediction control; when intermediate mechanisms are intervened on, a flat predictor without action inputs cannot represent that response using roots alone. Such rows are excluded from its observational-map fit.

The connection between neural models and SCMs already has substantial theory. Expressiveness is distinct from causal learnability, and structural constraints and identification assumptions matter (Xia et al., 2021). We do not claim that neural approximation or low MSE identifies an unrestricted SCM. Our conditional mechanism regression and deterministic propagation statements address a much narrower known-graph setting. Causal Chambers provide an external physical apparatus with published causal models (Gamella et al., 2025); the archive study assesses physical prediction under that apparatus’s conditions, without claiming new live experiment design.

3 Completed delivery confirmation

The comparison retains the originally registered 12 histories of a single deterministic emulator. The delivery procedure fits separate mechanism MLPs from all eligible paid rows using full-batch Adam and additional epochs. Original online weights remain unchanged. Evaluation runs the learned chain on the full previously exposed discrete grid using predicted intermediate values. Acquired input points occur within this grid, so it is not a held-out prospective sample.

The primary endpoint is exact-level error after mapping predictions to the nearest target level. For each history, the registered numerator is the median of three scored optimization initializations, and the denominator is the original online chain’s error. That median summarizes sensitivity; it is not a deployable rule for selecting a model without access to true test outcomes. Histories, not initialization repeats or grid rows, are the paired units.

The geometric mean paired ratio is 0.183, with a log-scale t interval of [0.102, 0.327] and exhaustive two-sided sign-flip $p = 0.00146484$. 11 of 12 histories improve. The worsening history 124753321 changes from 0.196339 online error to 0.395658 delivery median error (ratio 2.015177); it is retained without exclusion or rescue tuning. The generated values and source hashes are listed in `claim_index.json`.

This supports the bundled delivery procedure on repeated histories of that emulator. Paid histories are matched, but fitting compute is not. It establishes neither acquisition superiority, causal-factorization attribution, a foundation model benefit, nor generalization over independently parameterized systems.

4 Attribution and prospective evaluation protocol

Stage A: attribution (exploratory; completed). We compare three unique-row data sets: the final 50-entry buffer, all rows actually admitted at live online updates, and all eligible paid rows. Each is fit from scratch for 100 or 30,000 epochs with fixed architecture, initialization scheme,

optimizer and intervention masks. The original online model is a separate comparator. Flat MLPs have both equal-epoch and separately labeled matched-CPU comparisons. Fixed ridge, polynomial and tree controls are configured on an earlier development history, with no configuration tuning on the twelve exposed histories.

Replay reconstruction reveals additional differences from the online training path: the selected observation is duplicated in consolidation; per-node fast adaptation can include a directly intervened node; and the final observation refresh enters the saved buffer after the last fit. These differences are recorded rather than silently attributed to data quantity. The scratch factorial controls masks among its cells but does not exactly replay the online objective. No selected clamped-node fast updates occurred in these archived confirmation histories, so that code-path limitation is not an observed explanation of their delivery improvement. Diagnostics retain support coverage, observed-parent residuals, propagated prediction shifts and decision margins, especially for the worsening history. No simulator responses are acquired in Stage A.

Attribution findings. The full registered matrix completed and passed an independent audit of source, inputs, eligibility masks, original artifacts, fitted-model hashes and all score files. With initialization zero, all-paid SCM fitting for 30,000 epochs gives geometric mean continuous NMSE ratios of 0.522 versus unchanged online weights, 0.551 versus the development-selected flat refit, and 0.423 versus the separately matched-CPU flat fit. These are descriptive results from the previously exposed grid, not prospective superiority tests.

Within the scratch SCM factorial, the all-paid long fit has ratios of 0.096 versus final-buffer long fitting and 0.034 versus all-paid 100-epoch fitting. Its ratio versus online-admitted long fitting is 0.980: this comparison supplies little evidence of a benefit from the previously unused paid observations. Thus retaining a completed history and allocating fitting effort are much larger explanations here than adding those observations. The registered exploratory rule selects the all-paid SCM at initialization zero, the frozen flat comparator and the short-fit optimization ablation for prospective evaluation.

SCM and flat fits share the paid history but differ in which intermediate-clamp rows can enter their regressions. The structured recipe can reuse measurements that a roots-only observational predictor cannot represent. Accordingly this comparison does not isolate a pure architecture effect with identical usable rows, nor prove general causal identifiability. Matched fit CPU includes every SCM head, including the communication-cost branch that does not feed the scored target. The complete matrix consumed 10.87 fit CPU-core-hours; this excludes process startup, imports and evaluation. Table 1 retains all primary configurations; initialization sensitivity is preserved separately.

Retained failure and propagation. History 124753321 still worsens under the primary delivery fit, with snapped error 0.397. Its target mechanism’s observed-parent MSE is 8.07×10^{-5} , while free-running chain MSE is 2.86×10^{-4} ; the propagated prediction-shift MSE is 2.09×10^{-4} . Only 0.020% of evaluated target parent values lie outside the training bounding box. This points to an in-box propagation and decision-margin failure rather than widespread box extrapolation, although a bounding box is not a support-density certificate and shifted predictions do not add independently to local residual MSE. The history is retained without tuning a rescue or selecting a different scored initialization.

Stage B: controlled generalization (gated; pending). After attribution, freeze a deployable recipe with initialization zero, the strongest simpler refit and a decisive ablation. Compare them and

Data	Learner	Epochs	NMSE/online	Snap/online	CPU h
Final50	SCM	100	26.889	4.082	0.002
Final50	SCM	30,000	5.466	2.593	0.390
Final50	flat	100	29.230	4.126	< 0.001
Final50	flat	30,000	29.158	4.103	0.078
Admitted	SCM	100	16.661	3.865	0.003
Admitted	SCM	30,000	0.533	0.435	0.783
Admitted	flat	100	13.198	3.695	0.001
Admitted	flat	30,000	2.395	1.949	0.159
All paid	SCM	100	15.448	3.846	0.006
All paid	SCM	30,000	0.522	0.204	1.455
All paid	flat	100	12.248	3.655	0.002
All paid	flat	30,000	0.948	0.950	0.270
All paid	ridge	–	28.890	4.150	0.002
All paid	polynomial	–	4.440	2.737	0.002
All paid	tree	–	24.582	3.883	0.004
All paid	flat	CPU	1.234	1.206	1.455

Table 1: Exploratory primary initialization0 across the twelve archived histories. Ratios are geometric means relative to unchanged online weights; lower is better. CPU hours sum fit process time across the twelve fits in each row, excluding imports and evaluation. SCM cost includes all five heads. CPU denotes the matched-CPU flat fit, whose update count varies. Epoch counts apply only to neural fits. All configurations and sensitivity initializations are retained in the receipt.

the online learner on independently parameterized five-node and thirty-node systems in the audited generator families. Both balanced varied-value and matched random histories are shared across prediction methods. Complete joint-action blocks are reserved before collection. Continuous held-out normalized MSE is primary; snapped error is secondary only for genuinely quantized outputs. Histories and optimization repeats are nested within systems. Paired log-ratio contrasts, multiple-testing correction, practical thresholds and separate graph-size reporting are frozen before opening outcomes. The stochastic estimand must be specified explicitly: composing conditional means through nonlinear downstream functions does not generally produce their interventional expected outcome. The primary protocol proposal disables noise during both collection and evaluation, targeting the same deterministic structural map for every arm. This setting is labeled as a noise-disabled parameterization of the audited families; it does not establish a benefit for stochastic interventional means. Each of 20 systems per graph-size stratum supplies two 400-row training histories and one shared 400-action held-out evaluation. All actions jointly clamp roots; the target is X_3 for five nodes and X_{30} for thirty. The first 50 paid training rows supply fixed shared calibration before the retrospective online updates; target variance for NMSE uses training data only. All 640 fits must be sealed before any held-out response is generated. Initialization zero is primary; initializations one and two measure sensitivity without scored model selection. Four primary tests (two controls in each stratum) use Holm correction and system-level paired contrasts. SCM fits use measured intermediate labels and train all nonroot heads; the flat fit receives roots and the target. Thus supervision and fitting compute differ, and the comparison tests a delivery recipe rather than a pure architecture effect. Root-only histories can leave joint parent support on a manifold (Section 5); this protocol does not test identification under arbitrary internal interventions.

Stage C: archived physical prediction (completed). We use the eleven previously unscored conditions of `lt_malus_v1`; `white_64` remains development only. Duplicate commands and joint

action blocks are grouped before splitting. Compare delivery, rolling-buffer learning, joint Fourier regression and the published relative-angle physics model, using continuous prediction error. The physics basis is an intercept and $\cos^2(\theta_1 - \theta_2)$, with degree inputs converted to radians; the earlier product-form attenuation control is not retained. Conditions and repeated readings from one apparatus are not independent physical worlds. All conditions and conditional uncertainty are reported in Tables 2 and 3. At fixed initialization zero, delivery has lower row-weighted NMSE than rolling-buffer fitting in 7 conditions, than the physics fit in 3, and than Fourier regression in 0. These are descriptive counts, not independent-world tests. The study consumed 0.053 fit/evaluation CPU-core-hours, excluding imports and startup, and acquired no physical responses. All fits were sealed before held-out losses were computed. Fourier regression outperforming delivery in every condition rules out an external advantage over that control here; the fixed domain-informed control provides better predictions. This case contains one measured response mechanism, so it is not a test of DAG factorization. We retain the result without changing the recipe after scoring.

Condition	Delivery	Rolling	Physics	Fourier
blue_128	0.359	0.156	0.058	0.049
blue_255	0.012	0.017	0.026	9.43×10^{-4}
blue_64	0.471	2.087	0.103	0.102
green_128	0.789	0.290	0.058	0.046
green_255	0.010	0.038	0.018	0.002
green_64	5.167	3.307	0.121	0.102
red_128	0.276	0.389	0.047	0.045
red_255	0.011	0.063	0.024	0.004
red_64	1.119	0.823	0.159	0.135
white_128	0.163	0.462	0.044	0.042
white_255	0.020	0.023	0.006	0.002

Table 2: Continuous held-out MSE divided by the training variance for every physical condition. Lower is better. These conditions are readings of one apparatus, not independent systems.

Condition	Delivery/Rolling	Delivery/Physics	Delivery/Fourier
blue_128	2.55 [0.50, 8.03]	6.61 [0.97, 17.08]	7.88 [1.40, 17.91]
blue_255	0.71 [0.24, 1.01]	0.47 [0.10, 1.22]	13.05 [2.29, 28.48]
blue_64	0.22 [0.07, 1.12]	4.38 [1.56, 8.49]	4.41 [1.56, 8.29]
green_128	2.58 [0.43, 4.38]	12.96 [1.15, 38.54]	16.22 [1.91, 46.75]
green_255	0.23 [0.06, 2.40]	0.54 [0.10, 1.57]	6.26 [1.41, 19.05]
green_64	1.56 [0.17, 1.78]	43.05 [1.14, 125.29]	51.08 [1.51, 136.39]
red_128	0.75 [0.08, 2.97]	6.47 [1.04, 14.30]	6.64 [1.21, 13.20]
red_255	0.18 [0.06, 0.60]	0.52 [0.18, 1.36]	2.74 [1.48, 4.21]
red_64	1.31 [0.19, 4.59]	6.54 [1.40, 16.18]	7.37 [1.73, 18.01]
white_128	0.36 [0.12, 1.12]	3.86 [1.09, 7.31]	3.95 [1.16, 7.86]
white_255	0.85 [0.16, 3.55]	2.92 [0.54, 6.83]	8.57 [1.65, 15.65]

Table 3: Equal-action-block error ratios and conditional 95% bootstrap intervals. Eight held-out blocks are resampled within each condition. These ratios weight blocks equally, whereas Table 2 weights held-out rows equally. Intervals describe uncertainty conditional on this apparatus and observed blocks; they are not independent-world confidence intervals or multiplicity-adjusted superiority tests.

5 Conditions and limits of delivery

We study a known acyclic graph with measured parents. The following statements explain permissible reuse and possible error amplification. They do not prove identification for arbitrary neural models or certify the fitted networks.

Proposition 5.1 (Mechanism eligibility). *Let $V_i = f_i(V_{\text{Pa}(i)}) + U_i$ be an invariant additive-noise mechanism. In each experiment, suppose the current exogenous noises are independent across mechanisms and independent of all past experiments; U_i is integrable and has mean zero. An intervention policy may depend on previous experiments, but must choose its action before the current exogenous noises are realized. Retain a row only when the action does not replace mechanism i , and when retention does not select on its current noise or response conditional on its parents. Then, in the retained interventional mixture,*

$$\mathbb{E}[V_i \mid V_{\text{Pa}(i)} = p, \text{retained}] = f_i(p).$$

Thus squared-loss regression can target this mechanism on supported parent values, assuming finite second moments. This is an estimand statement; finite data, optimization and extrapolation guarantees require further assumptions.

Proof. In an acyclic graph, a node’s parents depend only on earlier exogenous noises and the chosen action. The current U_i is independent of those noises and of past experiments. Since the policy precedes the current realization, and the retention rule does not condition on U_i , the conditional expectation of U_i remains zero. Substituting the structural equation yields the assertion. The conditional expectation minimizes population squared loss on the retained support. No statement outside that support follows. $\square$

Confounding, temporal dependence of current noise on prior experiments, missing parents, interventions that change rather than clamp a mechanism, and response-dependent retention can invalidate the premise. In particular, selecting the best probe using its measured response can change the noise distribution. All-paid reuse avoids discarding probes on that basis, but still requires invariant mechanisms and measured parents. The current confirmation uses a deterministic emulator; this distinction matters for any extension to noisy systems. A non-additive stochastic SCM also needs a distributional treatment: composing conditional means need not produce a downstream interventional mean through a nonlinear mechanism. For example, clamp $W = 0$, set $X = W + U$ with $\Pr(U = 1) = \Pr(U = -1) = 1/2$, and let $Y = X^2$. The conditional-mean mechanism for X gives zero at $W = 0$, so composing the mean mechanisms predicts zero for Y . The actual interventional expectation is one. The zero-noise structural map and the noisy interventional mean are distinct estimands, even when both conditional-mean mechanisms are exactly known.

Proposition 5.2 (Deterministic DAG propagation). *Consider a deterministic SCM and its learned functions $\hat{f}_i$, evaluated in topological order under identical correctly clamped interventions. For each non-clamped node, assume $|\hat{f}_i(v_{\text{Pa}(i)}) - f_i(v_{\text{Pa}(i)})| \leq \epsilon_i$ at every true parent vector in the deployment support. Assume also*

$$|\hat{f}_i(p) - \hat{f}_i(q)| \leq \sum_{j \in \text{Pa}(i)} L_{ij} |p_j - q_j|$$

on a region containing both true and predicted parent vectors, with $L_{ij} \geq 0$. Writing $e_i = |\hat{v}_i - v_i|$, one has

$$e_i \leq \epsilon_i + \sum_{j \in \text{Pa}(i)} L_{ij} e_j.$$

Correctly clamped nodes have $e_i = 0$. If their rows in L and their local bounds are set to zero, the vector bound is $e \leq (I - L)^{-1}\epsilon = \sum_{k=0}^{d-1} L^k \epsilon$, where d is the number of nodes and L is strictly triangular in topological order.

Proof. Add and subtract $\hat{f}_i(v_{\text{Pa}(i)})$ and apply the triangle inequality. The local approximation bounds one term; the coordinate Lipschitz inequality bounds the other. Clamps are identical by assumption. Substitution in topological order gives the path sum. A strictly triangular $d \times d$ matrix satisfies $L^d = 0$, so the finite geometric sum is its inverse. $\square$

The bound can amplify a small local error along deep or high-gain paths. Observed-parent test MSE is neither a uniform ϵ_i nor a verified Lipschitz constant. Our residual and support diagnostics are explanatory measurements, not certified instances of this proposition.

Proposition 5.3 (Root interventions need not constrain propagation). *Consider the known graph $R \rightarrow M$, $R \rightarrow Y$, $M \rightarrow Y$, with $M = R$ and $Y = f(R, M) = R + M$. For any $K \notin \{-1, 1\}$, the alternative mechanism $g_K(R, M) = R + M + K(M - R)$ uses both declared parents and agrees with f under every intervention on R alone. If a learned upstream mechanism instead predicts $\hat{M} = R + \delta$, its free-running prediction using g_K has error $|(1 + K)\delta|$, although its observed-parent fit is exact.*

Proof. Root interventions leave the parent support of Y on the diagonal $M = R$, where the added term vanishes. Substitution of $\hat{M}$ gives $g_K(R, \hat{M}) = 2R + (1 + K)\delta$, whereas the true outcome is $2R$. Both parent coefficients are nonzero for the stated choices of K . $\square$

This failure need not leave a coordinate bounding box. For example, take $R = 0.25$, $\delta = 0.01$ and $K = 1000$. Both true and predicted parent vectors are inside $[-1, 1]^2$, but the prediction error is 10.01. Observations covering the diagonal cannot constrain sensitivity normal to it; more root observations alone do not remove that ambiguity. An internal intervention can distinguish these mechanisms. Our prospective root-only protocol assesses prediction under held-out root interventions, not unrestricted internal-action identification. This is an existence example, not a diagnosis of a measured failure or a verified bound for the fitted networks.

Proposition 5.4 (Quantization margin). *For distinct ordered levels $\ell_1 < \dots < \ell_m$, let Q assign the nearest level, with a fixed tie rule, and let $b_k = (\ell_k + \ell_{k+1})/2$. Define $M(y) = \min_k |y - b_k|$. If $|\hat{y} - y| < M(y)$ then $Q(\hat{y}) = Q(y)$. If a pointwise prediction bound $B(y)$ is valid, $\Pr(Q(\hat{y}) \neq Q(y)) \leq \Pr(M(y) \leq B(y))$.*

Proof. The open interval of radius $M(y)$ around y crosses no decision boundary. Consequently both values occupy the same decision cell. The contrapositive implies that a quantization error requires $M(y) \leq |\hat{y} - y| \leq B(y)$. Strict inequality excludes boundary ties. $\square$

Snapped error depends on the location of errors relative to decision boundaries; similar MSE alone need not imply similar exact-level accuracy.

Counterexamples. For a constant predictor and test response zero, fitting a single training response zero gives zero risk; adding a noisy response two makes its least-squares mean one and test risk one. With a noisy training target one, optimizing a constant from 0.1 to one reduces training error but increases test error. For a two-stage chain, suppose the true parent and child are zero on the observed support, the learned parent predicts 0.1, and the learned child is $\hat{f}(z) = 100z$. The child has zero observed-parent error at zero, but its free-running prediction is ten. These are existence examples, not descriptions of the measured histories. Analytically solved chains, forks, colliders and clamps are checked by executable tests.

6 Reproducibility and resource accounting

Fitting and evaluation run in separate processes. Training inputs, row selections, source, dependency versions, optimizer configuration and outputs are hash-bound. Stage A fit workers are denied network access and evaluation arrays. The physical worker uses only registered training rows for fitting and normalization and seals predictions before computing held-out losses. Exclusive output paths preserve interrupted attempts and reject replacement. Supervisors enforce wall-time and memory ceilings. CPU time, wall time, unique eligible rows, parameters, optimizer updates and memory accompany each fit. Original histories and weights are retained under local result custody. The claim generator binds every retained numerical result to its validated receipt. Experimental controls, analytic checks and the execution ledger are versioned with the manuscript. No closed-source API is needed for the critical path.

7 Limitations and conclusion

The current positive confirmation is restricted to a bundled delivery recipe, a quantized endpoint, and repeated histories of a single fixed emulator on an exposed evaluation grid. Exploratory continuous-error attribution is favorable, but physical prediction loses to the Fourier control in every condition. The evidence does not yet justify a new architecture or equal- compute superiority. The theoretical statements require graph, invariance, noise and support assumptions that cannot be inferred from small fitted residuals. Additional observations and optimization can worsen test performance, and local accuracy can fail under chain deployment.

Submission requires an attributable benefit, prospective controls, an honest external boundary, compute accounting and complete reproducibility. If ordinary refitting or matched compute removes the distinction, a narrower empirical methodology report is the appropriate outcome. The present draft makes those unresolved requirements explicit.

References

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